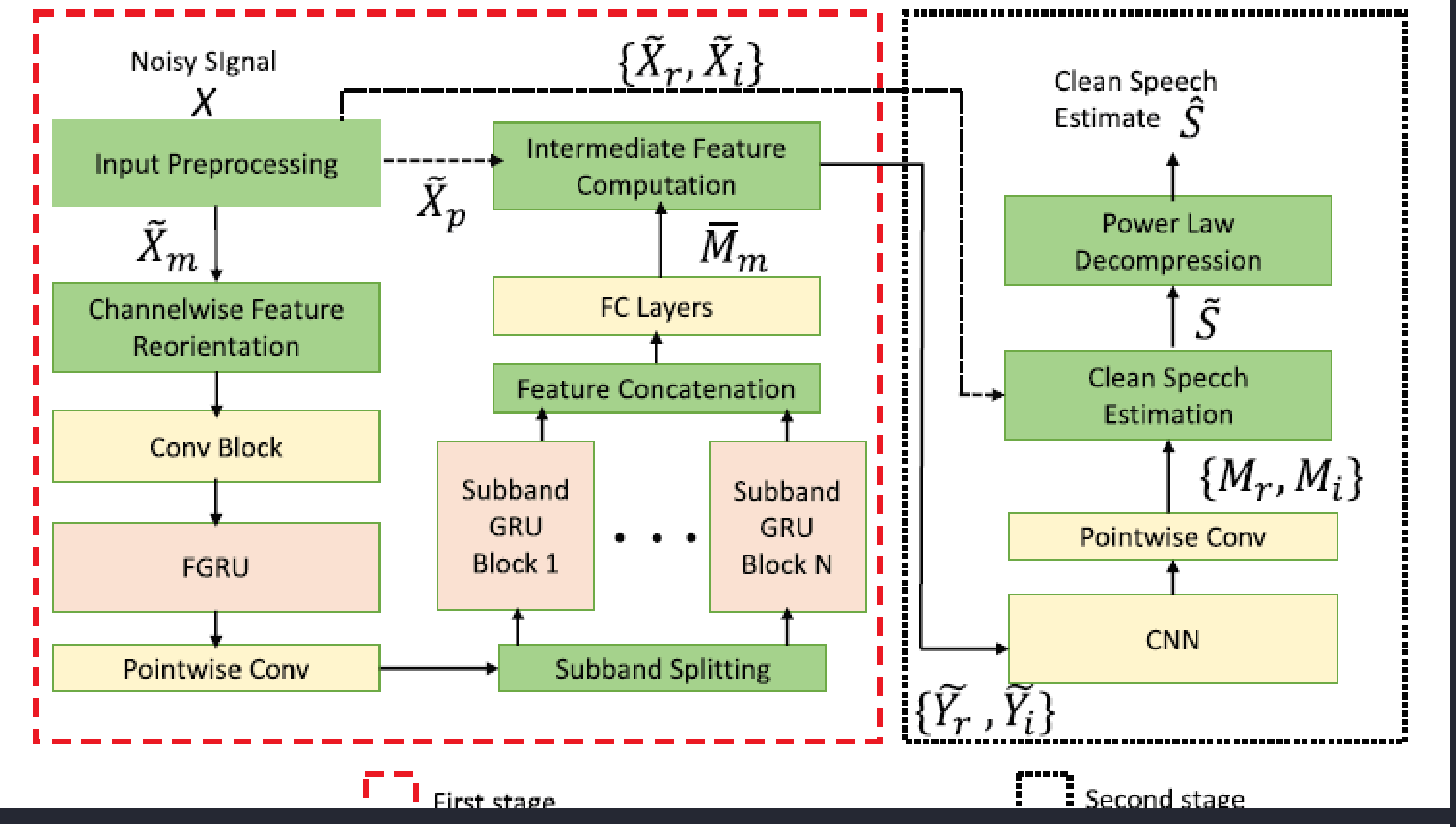
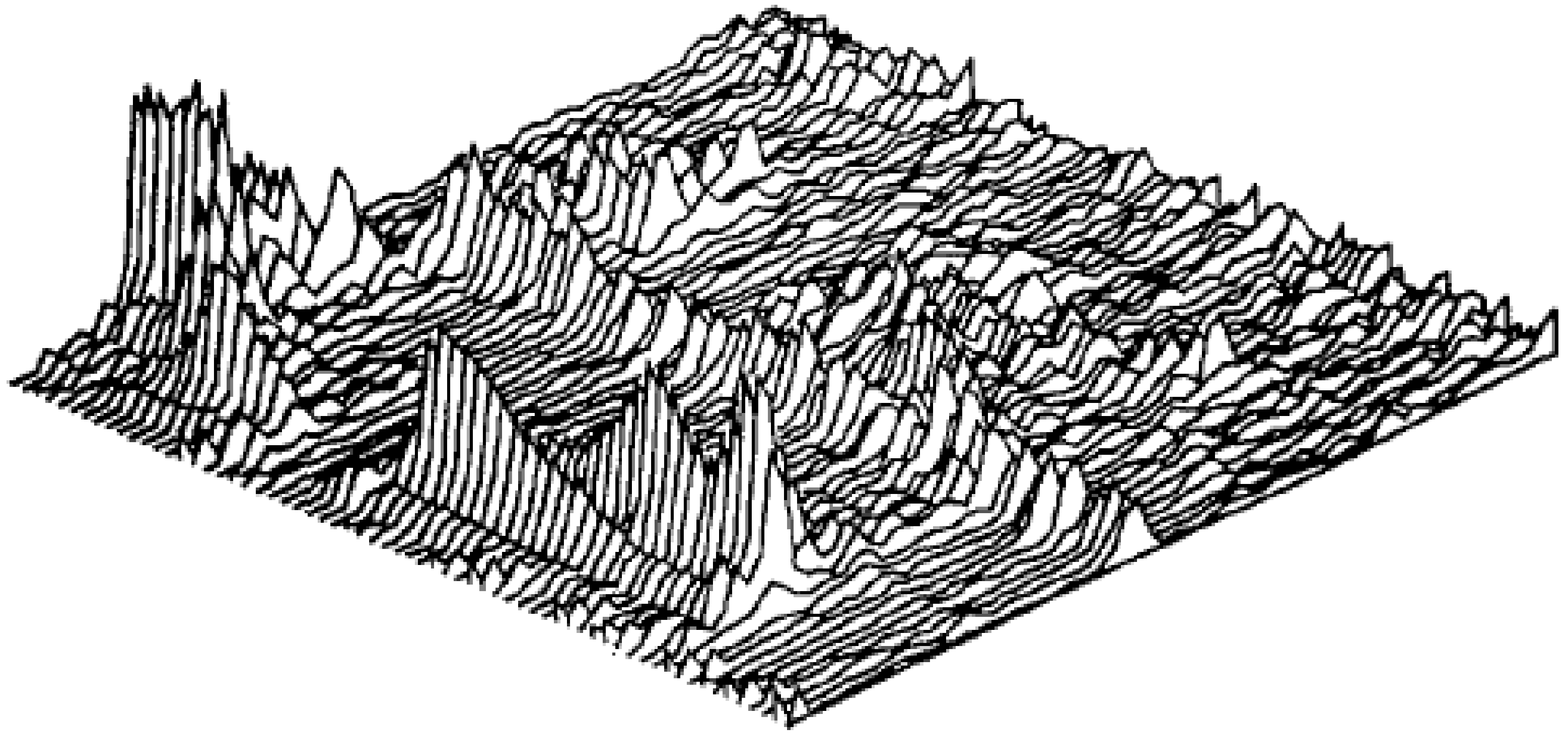




Comparison of Classical and Deep Learning Approaches for Speech Noise Suppression

Aryan Gupta (230150003)

Project Objective

This research compares classical STFT-based speech enhancement with deep learning noise suppression models, demonstrating performance differences through an interactive evaluation framework.

Motivation

The Challenge

Real-world speech signals are frequently degraded by background noise, affecting intelligibility, audio quality, and the performance of automatic speech recognition (ASR) and communication systems.

Two Paradigms

- **Classical signal processing**: Time-frequency domain operations using noise estimation and filtering
- **Deep learning methods**: Learned mappings from noisy to clean speech with complex noise pattern modeling

Research Goal: Experimentally evaluate the practical differences between these two approaches across multiple performance dimensions.

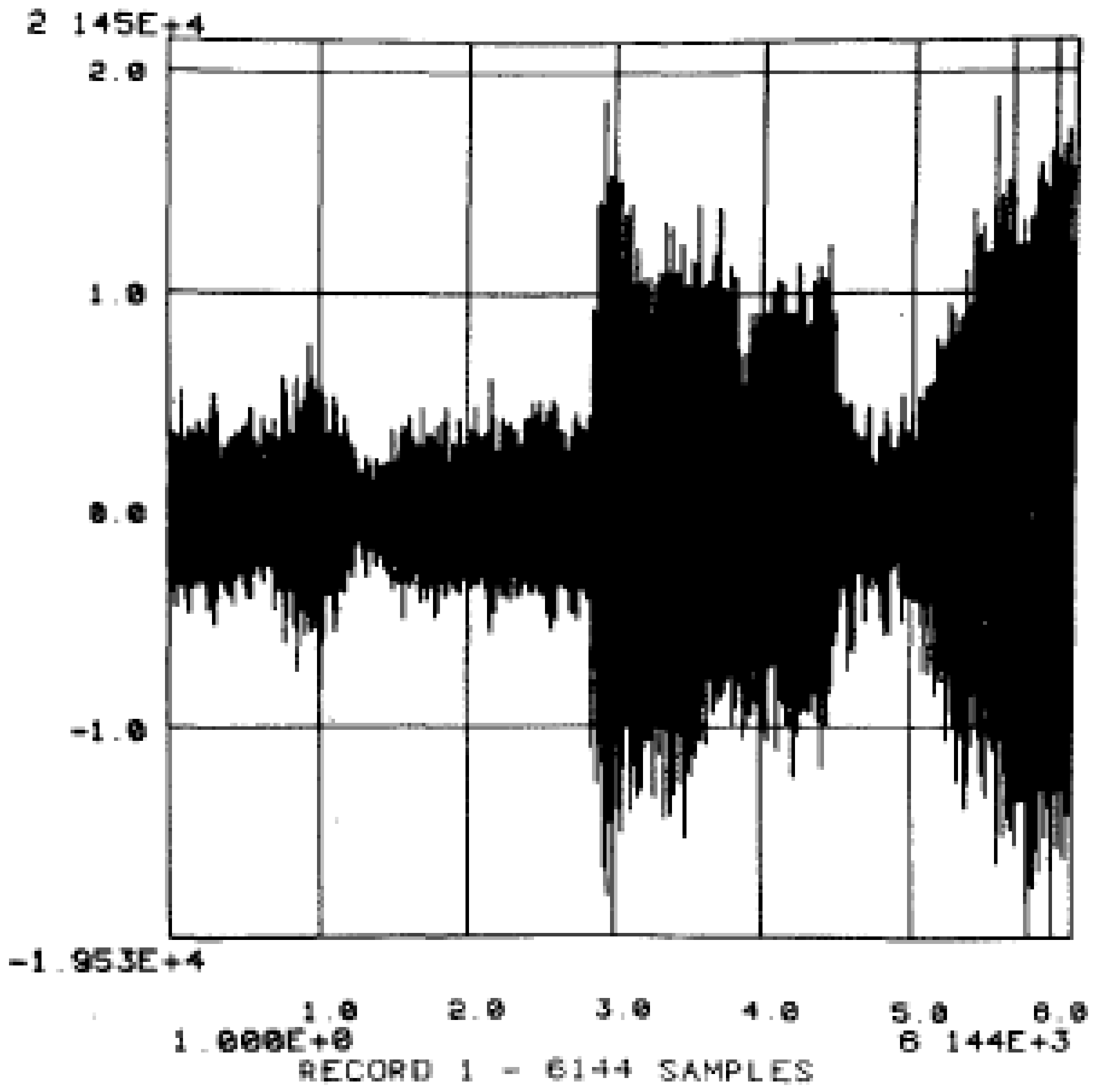


Fig. 4. Time waveform of helicopter speech. "Save your".

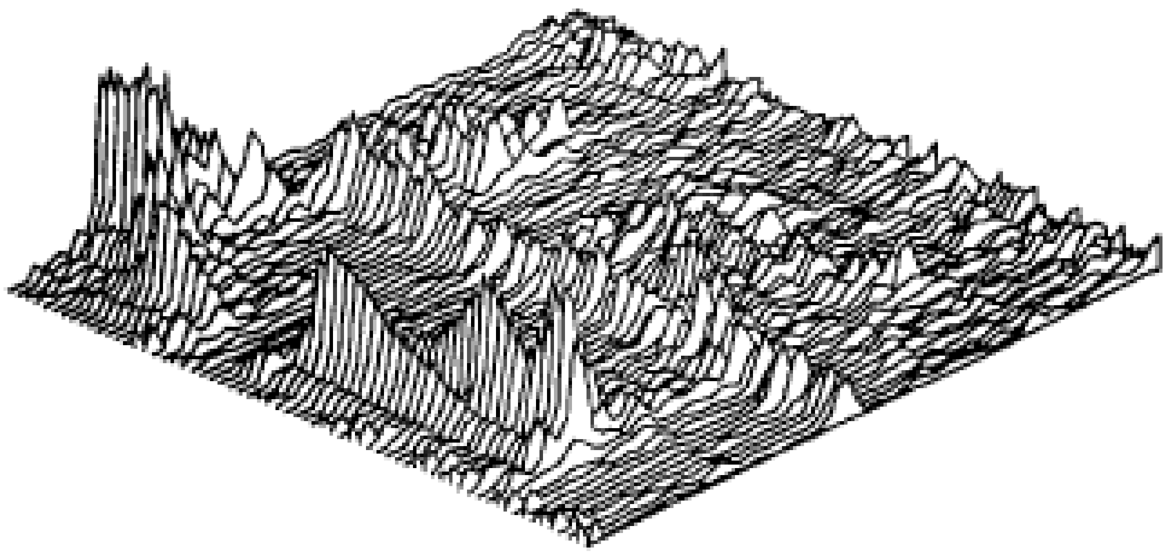


Fig. 6. Short-time spectrum of helicopter speech.

Dataset

Experiments utilize the “Reverberant speech database for training speech dereverberation algorithms and TTS models”, introduced by Cassia Valentini-Botinhao and Junichi Yamagishi.

Dataset Composition

Clean speech recordings from the VCTK corpus combined with noise recordings from the DEMAND dataset.

Training Set

56 speakers (28 male + 28 female) providing diverse vocal characteristics and speaking styles

Test Set

2 speakers reserved for unbiased performance evaluation

Noise Conditions

Multiple real-world noise types at signal-to-noise ratios 0-15 dB in train set and 2.5-17.5 db in test set

Dataset Sample - Noisy Datapoint

Dataset Sample - Corresponding Clean Datapoint

Classical Speech Enhancement Approach

Traditional STFT-Based Processing

Short-Time Fourier Transform (STFT) processing remains the foundation of classical speech enhancement. The well-known spectral subtraction method, proposed by Steven F. Boll, represents a cornerstone technique.

STFT Analysis

Convert noisy speech to frequency domain

Noise Estimation

Estimate noise spectrum characteristics

Spectral Suppression

Suppress noise in magnitude spectrum

Signal Reconstruction

Inverse STFT to recover waveform

Advantages

- Simple, computationally efficient
- Suitable for real-time applications

Limitations

- Sensitive to noise estimation errors
- May introduce artifacts

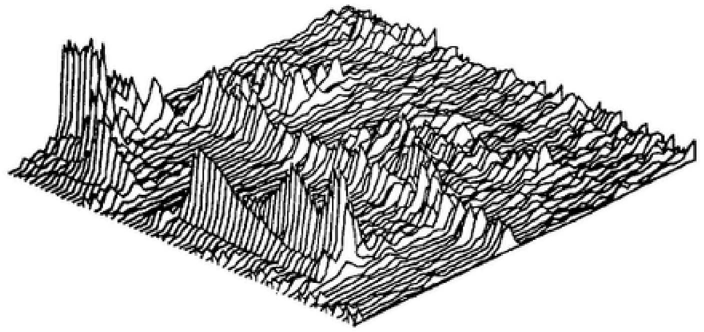


Fig. 6. Short-time spectrum of helicopter speech.

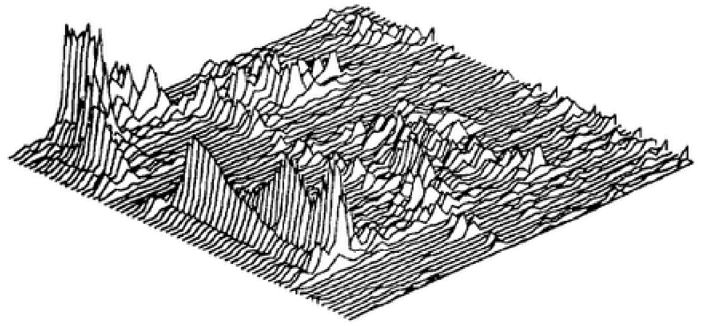


Fig. 7. Short-time spectrum using bias removal and half-wave rectification.

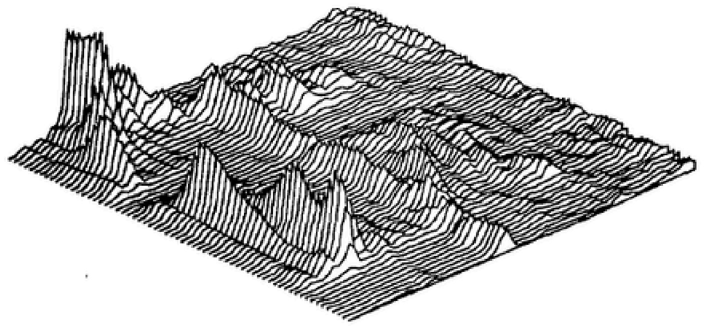


Fig. 8. Short-time spectrum of helicopter speech using three frame averaging.

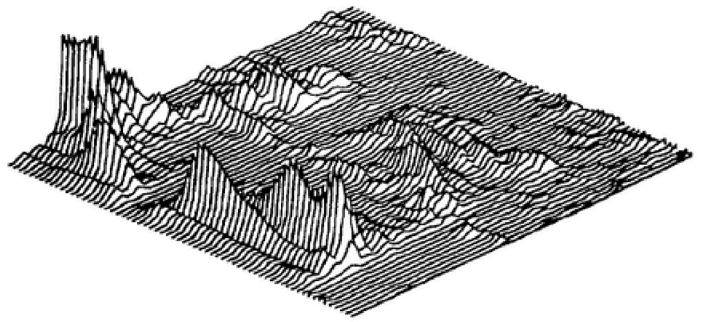


Fig. 9. Short-time spectrum using bias removal and half-wave rectification after three frame averaging.

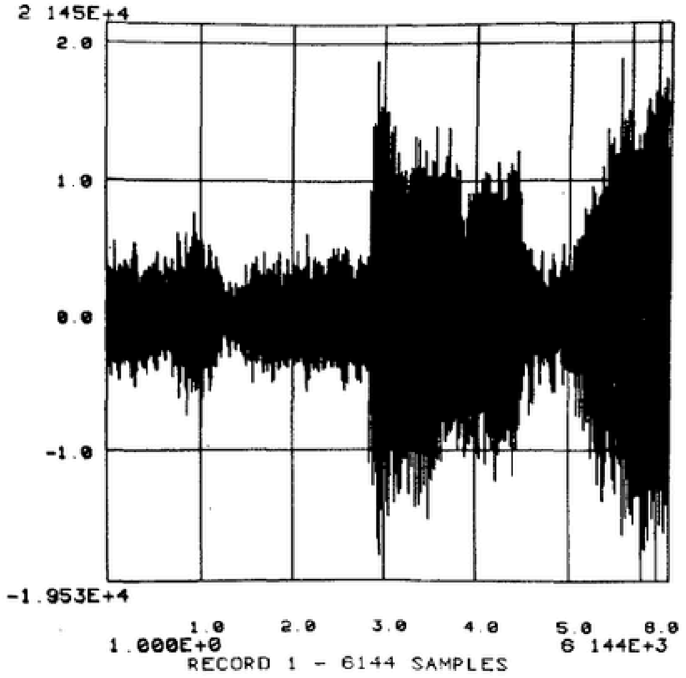


Fig. 4. Time waveform of helicopter speech. "Save your".

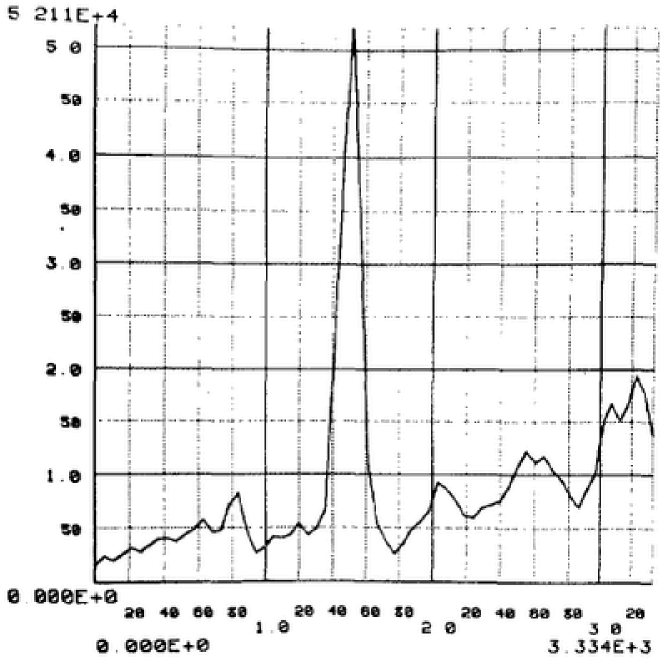
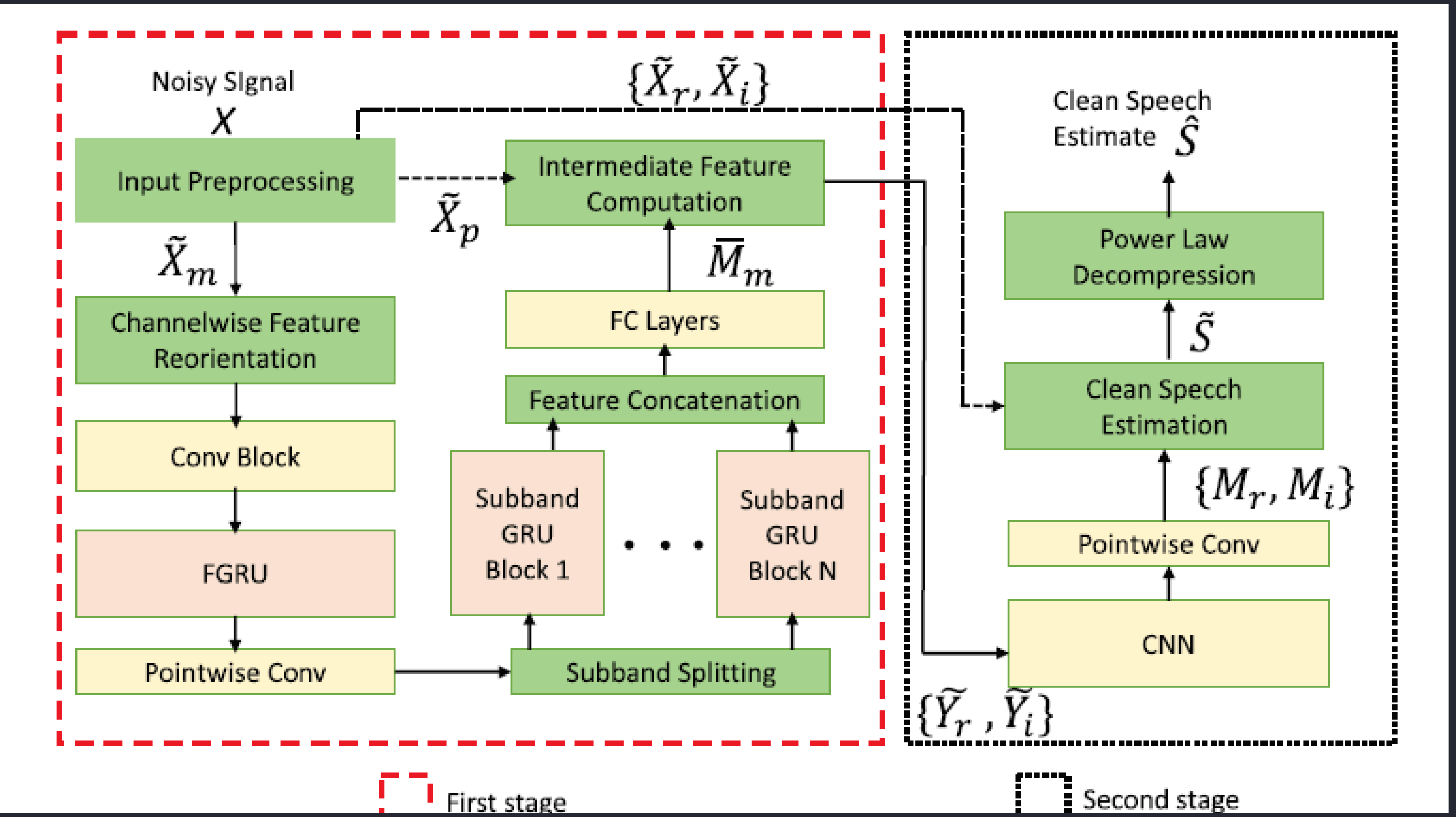


Fig. 5. Average noise magnitude of helicopter noise.

Deep Learning Based Approach

Neural Network Enhancement

Modern speech enhancement leverages neural networks to learn complex mappings from noisy speech to clean representations. This project employs a model inspired by ULCNet operating in the STFT domain.



01

STFT Transformation

Convert noisy speech to time-frequency representation

02

Power-Law Feature Compression

reduce dynamic range and improve learning stability

03

Intermediate Mask Estimation

Neural network predicts spectral mask to identify noise components

04

Complex Mask Refinement

small CNN refines the intermediate estimate and predicts a complex spectral mask to better reconstruct the speech signal

05

Waveform Reconstruction

Inverse STFT recovers enhanced time-domain signal

Improved Performance

Superior noise suppression across diverse noise conditions

Complex Pattern Modeling

Capable of learning intricate noise structures

Environmental Adaptability

Generalizes across different acoustic environments

Evaluation Metrics

Performance assessment employs established objective speech enhancement metrics spanning quality, intelligibility, and computational efficiency dimensions.

Mean Squared Error

MSE between the enhanced speech and the clean speech will give a good estimate of how the model performed.

Average SNR reduction

This will help compare the models. The better model will have more SNR reduction.

Interactive Demonstration

Gradio Web Interface

An interactive application enables users to directly compare both approaches through hands-on experimentation with their own audio samples.

Audio Upload

Users can upload custom noisy speech samples for evaluation

Parallel Processing

Both enhancement methods process the same input simultaneously

Audio Playback

Listen to enhanced outputs from classical and deep learning approaches side-by-side

Metric Display

View quantitative evaluation metrics for objective performance comparison

Thank You

~Aryan Gupta (230150003)